

★ WEEK 10 · PART 3 · PRACTICE

Regimes in Action, Worked by Hand

Choose the Number of States; Gate the Trade by Regime

Model selection by BIC, and Stage IV of the pairs strategy

Part 3's practice, with the arithmetic shown. We pick the number of regimes with a penalized likelihood, then put Stage IV to work - using the filtered regime probability to gate the pairs trade and step aside in a crisis.

3P

PART 3 · APPLICATIONS

Choose the Model, Then Gate the Trade

Case A: compare one-, two-, and three-state HMMs by BIC and let the penalty pick two. Case B: gate the pairs trade on the crisis probability and compare the P&L with and without the gate. Then a Claude Code lab.

What Part 3 Told Us

Now we make it concrete - with numbers

1

How many regimes is a choice

Not a fact of nature. A penalized likelihood like BIC balances fit against complexity - and usually lands on two.

2

Turn a fitted HMM into a live read

Use the filtered probability, not the smoothed one, to avoid look-ahead. It's the honest, real-time regime estimate.

3

Stage IV: trade only when it's safe

Cointegration breaks in crises, so the regime becomes a gate. We compute the P&L it saves, by hand.

Everything here is a calculator exercise: three model scores, five trading days, one gate.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from fitted scores or a signal.
- Compute the BIC or the gated P&L by hand.
- Then read the model or trading decision.

What you should watch

- How the penalty stops you overfitting regimes.
- How the gate steps aside in a crisis.
- How Stage IV completes the pairs thread.

How Many Regimes?

Let a penalized likelihood decide

More regimes always fit the data better - but at some point you're just fitting noise. BIC scores each model by its fit minus a penalty for its parameters. We compute it for one-, two-, and three-state HMMs and let the lowest score pick the model.

The trade-off

- More states raise the log-likelihood.
- But each state adds parameters to penalize.
- BIC balances the two.

What we'll do by hand

- Count parameters for each model.
- Compute BIC for one, two, three states.
- Pick the lowest.

The BIC Criterion

Fit, minus a penalty for complexity

$$\text{BIC} = -2 \log L + k \ln n \quad (\text{fit reward, minus an overfitting penalty; lower is better})$$

The two terms

- Minus-two-log-L rewards a better fit.
- $k \ln(n)$ penalizes more parameters.
- Lower BIC is better overall.

Why it works

- A richer model must earn its extra parameters.
- If the fit gain is small, the penalty wins.
- It guards against overfitting regimes.

BIC for One, Two, Three States

Fit gains, penalty grows

$$\text{1-state : } -2(-820) + 2(6.22) = 1652$$

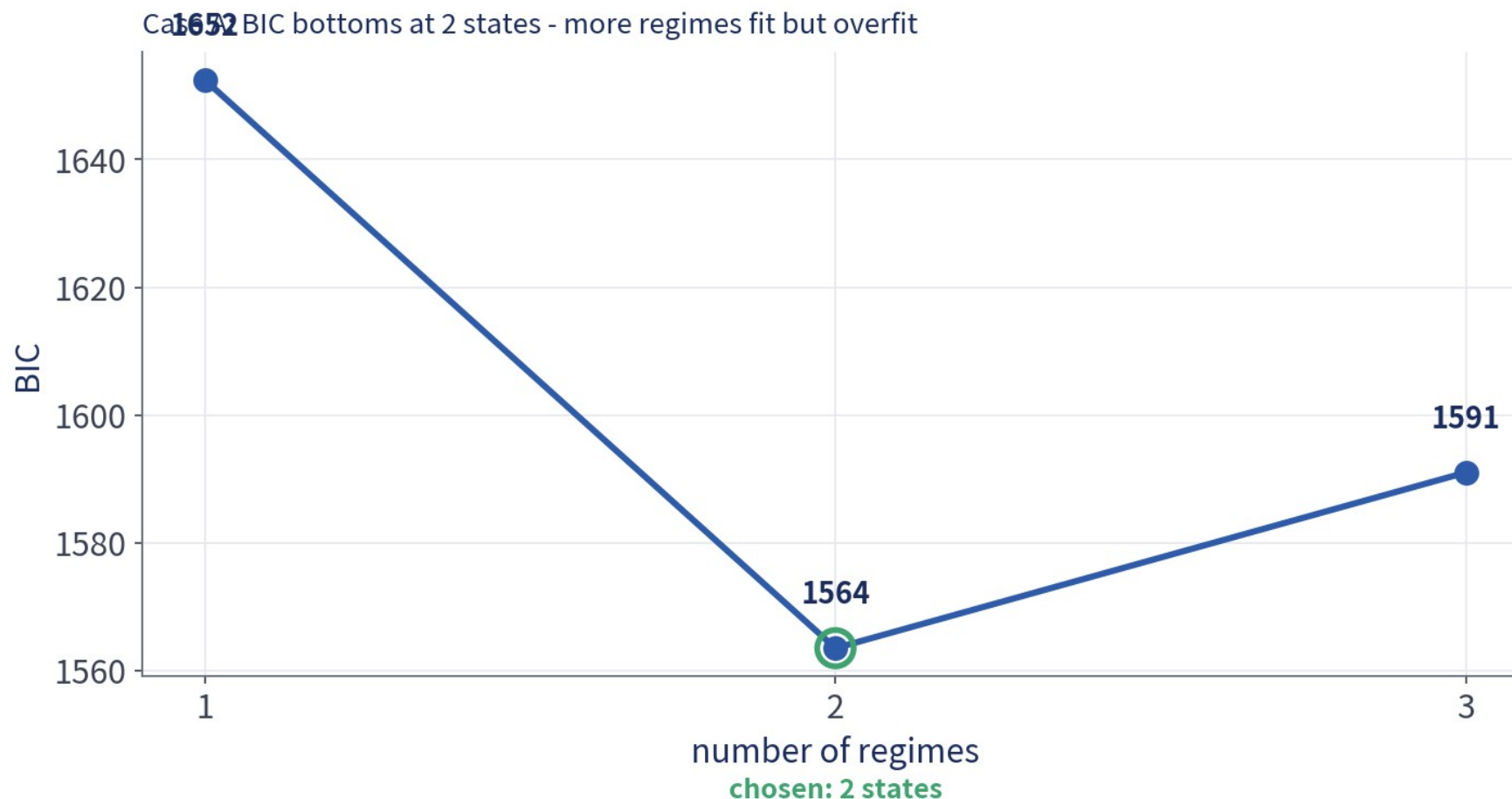
$$\text{2-state : } -2(-760) + 7(6.22) = \mathbf{1564} \quad (\text{lowest})$$

$$\text{3-state : } -2(-752) + 14(6.22) = 1591$$

Three states fit slightly better than two (log-L -752 vs -760), but the extra parameters cost more than they're worth - so the two-state model has the lowest BIC.

BIC Bottoms at Two States

The BIC falls sharply from one to two states as the fit improves, then rises again at three as the penalty for extra parameters overtakes the small gain. The curve's minimum - two states - is the calm/crisis default, chosen not by taste but by the score



Don't Overfit the Regimes

Two states is usually enough

The temptation

- More regimes always look better in-sample.
- You can label every wiggle its own state.
- It won't generalize.

The discipline

- Let BIC penalize the extra parameters.
- Two states - calm and crisis - is the default.
- Interpretable, robust, and usually right.

Stage IV: Gate the Trade

The thread learns when NOT to trade

Cointegration - the anchor of the pairs trade - breaks down in a crisis, exactly when the spread blows out. Stage IV uses the fitted HMM as a gate: trade the spread only when the crisis probability is low. We compute the P&L with and without the gate over five days.

The setup

- Five days of spread-trade P&L.
- The filtered crisis probability each day.
- Two of the days are a crisis.

What we'll do by hand

- Apply the gate: trade only if crisis is unlikely.
- Sum the gated and ungated P&L.
- See what the gate saved.

The Regime Gate

A position switch driven by the regime

$$\text{position}_t = \text{signal}_t \times \mathbf{1}[P(\text{crisis})_t < 0.5] \quad (\text{trade only in a favorable regime})$$

The rule

- Take the signal only in a calm regime.
- When crisis is likely, stand flat.
- The regime is an on/off switch.

Why filtered, not smoothed

- You must decide in real time.
- Smoothed uses future data - look-ahead.
- Use the filtered probability to be honest.

Five Days, Gated

Trade only when crisis is unlikely

day	spread P&L	P(crisis)	gate	gated P&L
1	+0.4	0.10	ON	+0.4
2	+0.3	0.15	ON	+0.3
3	-2.5	0.80	OFF	0.0
4	-1.8	0.70	OFF	0.0
5	+0.5	0.20	ON	+0.5

The gate is OFF on days 3 and 4, when the crisis probability is high - exactly the days the spread trade lost badly. Those losses are avoided entirely.

With and Without the Gate

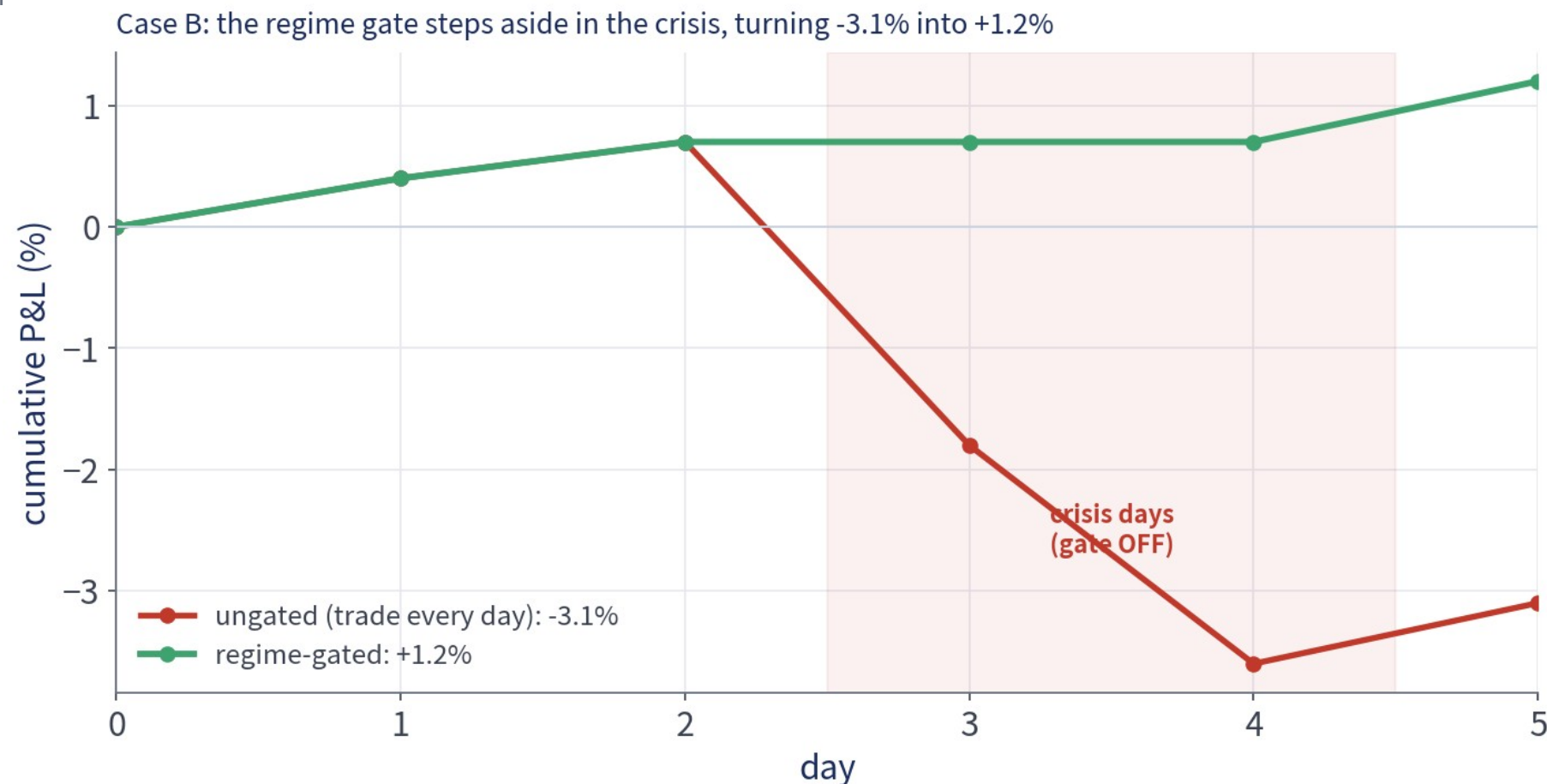
The P&L the regime saved

$$\begin{aligned}\text{ungated} &= 0.4 + 0.3 - 2.5 - 1.8 + 0.5 = -\mathbf{3.1\%} \\ \text{regime-gated} &= 0.4 + 0.3 + 0 + 0 + 0.5 = +\mathbf{1.2\%}\end{aligned}$$

Trading every day loses 3.1%; gating on the regime turns it into a 1.2% gain - a 4.3-point swing, purely from standing aside in the crisis.

The Gate Steps Aside in the Crisis

The ungated curve (red) collapses through the shaded crisis days as the spread blows out. The gated curve (green) simply goes flat there, having read the regime, and finishes positive. Stage IV's value is the loss it refuses to take



Hard Gate or Soft Sizing?

On/off, or scale by conviction

The hard gate

- Fully on below 0.5 crisis, fully off above.
- Simple, and it cleanly dodged the crisis.
- But it's sensitive to the threshold.

Soft conviction sizing

- Scale the position by $P(\text{calm})$ instead.
- Smoother, but keeps some crisis exposure.
- Here it netted about zero - the hard gate won.

Why Stage IV Matters

Knowing when not to trade

The danger it guards

- Cointegration breaks in a crisis.
- The spread blows out, not reverts.
- That's when the trade loses most.

The edge it adds

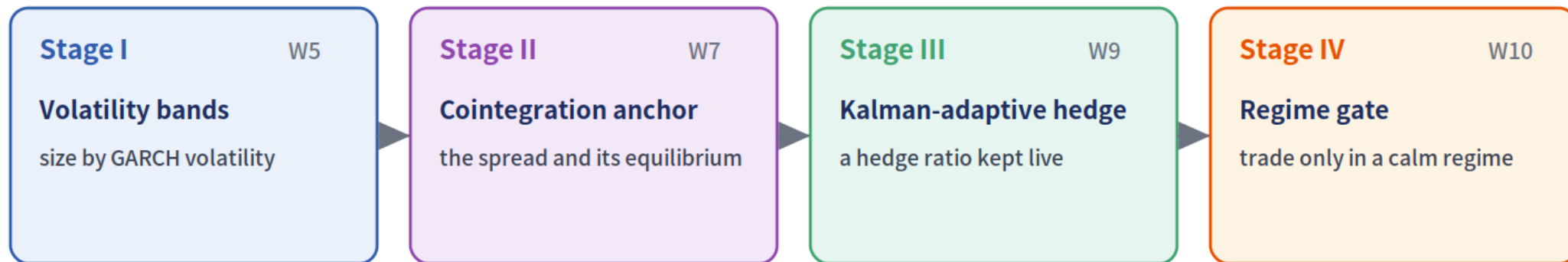
- The HMM sees the regime shift coming.
- It turns the trade off before the damage.
- Not trading is itself a decision.

The Pairs Journey, Now Four Stages

Each week added one piece: volatility bands size the trade (W5), cointegration anchors the spread (W7), a Kalman filter keeps the hedge live (W9), and now a regime gate decides whether to trade at all (W10). Stage IV is the discipline of standing aside.

The Pairs Journey, Now Four Stages

Each week added one piece; Stage IV teaches the strategy when NOT to trade.



Stage IV is the gate

Cointegration breaks in crises - so the HMM turns the trade OFF when the crisis regime is likely.

Reproduce the Selection and the Gate

Direct Claude Code to check your arithmetic

GOAL Verify the BIC choice and the regime-gated P&L, then backtest the gate.

[INPUT] Prompt · Claude Code

Write and run `part3_practice.py` that:

CASE A - model selection:

- * for $\log-L = -820, -760, -752$ and $n = 500$, compute k and BIC for 1, 2, 3 states; print the winner.

CASE B - the gate:

- * from the 5-day P&L and crisis probabilities, compute ungated and regime-gated totals;
- * also try soft sizing by $P(\text{calm})$.

Then fit a 2-state HMM on a real spread and backtest the gated strategy vs always-on.

Match every printed number to the slides.

Check against the slides

- BIC 1652, 1564, 1591.
- chosen: 2 states.
- ungated -3.1%, gated +1.2%.
- soft sizing ~ 0 .

Deliverable: `part3_practice.py` whose printed numbers match the BIC choice and the gated P&L.

Three Questions on the Cases

If you can answer these, the practice has landed

- Q1** Why not just pick the most regimes?
-
- Q2** Why gate the pairs trade on the regime?
-
- Q3** Filtered or smoothed probability for the gate?
-

BIC penalizes parameters - it overfits

Cointegration breaks in a crisis

Filtered - smoothed peeks at the future

All three come from the BIC selection and the Stage IV regime gate.

Three Pitfalls to Avoid

Where regime trading slips

Common mistakes

- "More regimes is a better model."
- "Use the smoothed regime to trade live."
- "A fitted HMM is a crystal ball."

The reality

- BIC penalizes complexity - two states usually wins.
- Smoothed uses the future - only filter is honest live.
- It's a probability, not a guarantee - size accordingly.

Two Calculations to Remember

Part 3 applications in one screen

1

Let BIC choose the number of regimes

Three states fit marginally better (log-L -752 vs -760), but with BIC 1591 versus 1564 the two-state model wins - the penalty for extra parameters outweighs the small fit gain. Calm and crisis, chosen by the score.

2

Gate the trade on the crisis probability

Trading every day lost 3.1% over five days; gating the trade off when crisis probability exceeded 0.5 skipped the two crisis days and turned it into a 1.2% gain - a 4.3-point swing from not trading.

3

Stage IV completes the pairs thread

Volatility bands, a cointegration anchor, a Kalman-adaptive hedge, and now a regime gate. The strategy has finally learned the hardest discipline: when to stand aside.

Next — Week 11 and continuous-time models: stochastic differential equations enter the story.